

SAMAR JAMAL

AI & Machine Learning Student

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Education

Graphic Era Hill University

B.Tech in Computer Science (Specialization in AI & ML)

Aug 2023 – Present

Dehradun, India

Oxford School of Excellence

Senior Secondary (PCM)

Apr 2022 – Mar 2023

Dehradun, India

Experience

FlyRank

AI/ML Intern

Aug 2026 – Present

Remote

Built a content-refresh ML workflow using anonymized search-performance data to prioritize pages for review.

Defined targets, features, data contracts, and validation; compared a gradient-boosting model using client-level holdout validation.

Evaluated Precision@K, Average Precision, and ROC-AUC; translated results into an actionable content-refresh playbook.

Amazon ML Summer School

Machine Learning Trainee

Jul 2026 – Present

Remote

Selected through a competitive, multi-stage nationwide process for a cohort trained by Amazon Applied Scientists and Engineers.

Built foundations in Python, PyTorch, deep learning, reinforcement learning, large language models, and causal inference through structured coursework and projects.

Projects

AI Disaster Response Coordinator | *OpenEnv Reinforcement Learning Environment*

Apr 2026 – Jun 2026

Built an OpenEnv-compliant RL environment in Python for disaster-response resource allocation, modeling priority triage and constrained dispatch.

Benchmarked a Qwen2.5-72B-Instruct baseline with a **0.884 weighted evaluation score**; containerized with Docker and served inference through FastAPI.

SenseLink | *Assistive IoT Platform for Accessibility*

Mar 2026 – May 2026

Built an edge-AI assistive system on ESP32 using YOLO for real-time object and obstacle detection with navigation support.

Integrated live captioning with on-device NLP and computer vision pipelines for constrained-device inference.

MediScan AI | *RAG-Powered Healthcare Intelligence Platform*

Nov 2025 – Jan 2026

Engineered a full-stack AI healthcare platform using React, TypeScript, Node.js, and PostgreSQL for disease prediction, drug recommendation, and heart-risk assessment.

Built a RAG chatbot with the Gemini API and Pinecone using semantic search and persistent conversational memory.

Designed contextual query rewriting and source-grounded retrieval to reduce ungrounded outputs and improve retrieval relevance.

Technical Skills

Programming Languages: Python, C/C++, Java

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, CUDA (AMP), NumPy, Pandas, Matplotlib

Generative AI & LLMs: Transformers, Hugging Face, Retrieval-Augmented Generation (RAG), Prompt Engineering

Computer Vision: YOLO, Real-Time Object Detection, Edge AI Inference

Backend & APIs: FastAPI, Django, REST APIs, Node.js

Web Development: React, TypeScript, HTML, CSS, JavaScript

Databases & Cloud: PostgreSQL, SQL, Supabase, Pinecone, AWS

Tools & Deployment: Git, Docker, Vercel, Jupyter, Colab

Certifications

Google Cloud Arcade Program – Completed 100+ labs and skill badges spanning BigQuery and AI/ML workflows *(2025)*

AWS Cloud Practitioner Essentials *(Sep – Oct 2025)*

Deep Learning and Reinforcement Learning, **IBM (Coursera)** *(Jan – Feb 2025)*

Extracurricular Activities

Top 75 Coder in **Amazon HackOn Season 6**, selected among thousands of participants nationwide.

2nd Runner-Up in **Hack-O-Holic 3.0** among 500+ participants.

Core Member, Codev Club – co-organized 3+ hackathons and coding events.

Active member, **IEEE Research Club** – contributed to technical events and innovation initiatives.